

**BACHELOR OF COMPUTER APPLICATIONS
(BCA) (Revised)**

Term-End Examination

June, 2021

**BCS-041 : FUNDAMENTALS OF COMPUTER
NETWORKS**

Time : 3 hours

Maximum Marks : 100

Note : Question no. 1 is **compulsory**. Attempt any **three** questions from the rest. Use of calculator is allowed.

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1. (a) Which type of communication, digital or analog, is better for computers ? Justify your answer. 5
 - (b) What are 'Hash functions' ? Why are they called 'One-way functions' ? Explain. 5
 - (c) Compare SVC and PVC of X.25 architecture. 5
 - (d) Why is serial data transmission faster than parallel data transmission ? Explain. 5

- (e) How are the number of twists in the UTP cable related to its performance ? 5
- (f) Name the two sublayers of Data Link Layer, and explain the characteristics of these layers. 5
- (g) Explain Silly Window Syndrome, with the help of a diagram. 5
- (h) Briefly discuss the problem with PSK and explain its possible solution. 5
- 2.** (a) Compare and contrast the amplitude modulation and phase modulation techniques. Discuss the limitations of each. 10
- (b) Explain RSA algorithm with example. 10
- 3.** (a) Explain the working of 3-way handshake used in TCP, using a suitable diagram. 10
- (b) Discuss the ATM service classes and QoS (Quality of Services) parameters of each class. 10
- 4.** (a) Calculate the CRC for a 10 bit sequence 1010011110 with a divisor of 1011.
Note : Show all steps and calculations. 10
- (b) Compare Time Division Multiplexing (TDM) and Frequency Division Multiplexing (FDM). Give the advantage and disadvantage (if any) of both TDM and FDM. 10

5. Write short notes on the following :

4×5=20

- (a) CSMA/CD
 - (b) IEEE 802.3
 - (c) Frame Relay
 - (d) OSI Model
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